

# Math 141

## Chapter 1.

§1.1.

Definition of function.

Vertical Line Test (Page 16)

Domain and Range of a function.

Piecewise function.

Symmetry of a function: Even function  $f(x) = f(x)$   
Odd function  $f(-x) = -f(x)$

Increasing/decreasing function.

§1.2.

Essential functions: polynomials, rational functions

Trigonometric functions, exponential functions

Logarithmic functions (domains and ranges).

§1.3.

Vertical and Horizontal shifts:  $y = f(x) \pm c$   
 $y = f(x \pm c)$

Vertical and Horizontal stretching:  $y = cf(x)$  or  $y = \frac{1}{c}f(x)$   
 $y = f(\frac{x}{c})$  or  $y = f(cx)$

reflecting:  $y = -f(x)$  or  $y = f(-x)$

Composite function:  $(f \circ g)(x) = f(g(x))$

domain of new function?

$f \pm g$ ,  $f * g$ ,  $\frac{f}{g}$

§1.4.

Viewing rectangle.

§1.5.

exponential function  $f(x) = a^x$

Law of exponents:  $a^{x+y} = a^x a^y$ ,  $a^{x-y} = \frac{a^x}{a^y}$ ,  $(a^x)^y = a^{xy}$ ,  $(ab)^x = a^x b^x$   
(Page 54)

The number  $e \approx 2.71828$  (Page 56).

§1.6.

One-to-One functions:  $f(x_1) \neq f(x_2)$  whenever  $x_1 \neq x_2$ .

↳ Horizontal Line Test (Page 60).

Inverse function:  $f^{-1}(y) = x \Leftrightarrow f(x) = y$ .

domain of  $f^{-1}$  = range of  $f$

range of  $f^{-1}$  = domain of  $f$

How to find the inverse function of a one-to-one function? (Page 62)

Logarithmic function:  $y = \log_a X \Leftrightarrow a^y = X$

Laws of Logarithms:  $\log_a(xy) = \log_a x + \log_a y$   
 $\log_a\left(\frac{x}{y}\right) = \log_a x - \log_a y$  (Page 64)  
 $\log_a(x^r) = r \log_a x$

Natural logarithm:  $\log_e X = \ln X$

$$y = \ln x \Leftrightarrow e^y = x$$

Inverse Trigonometric functions:  $y = \sin x \Leftrightarrow \sin^{-1} y = x$  or  $\arcsin y = x$   
 $y = \cos x \Leftrightarrow \cos^{-1} y = x$  or  $\arccos y = x$   
 $y = \tan x \Leftrightarrow \tan^{-1} y = x$  or  $\arctan y = x$   
(domains and ranges)

## Chapter 2

§2.1

Tangent line to a curve at a point.  
instant velocity

§2.2.

Definition of Limit of a function

$$\lim_{x \rightarrow a} f(x) = L \quad \text{ie. } f \rightarrow L \text{ as } x \rightarrow a. \quad (\text{Page 88}).$$

left-hand limit  $\lim_{x \rightarrow a^-} f(x) = L$

right-hand limit  $\lim_{x \rightarrow a^+} f(x) = L$

$$\lim_{x \rightarrow a} f(x) = L \quad \text{if and only if} \quad \lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = L.$$

Infinite limits

$$\lim_{x \rightarrow a} f(x) = +\infty \text{ or } -\infty.$$

$$\lim_{x \rightarrow 0} \frac{1}{x^2} = +\infty$$

$$\lim_{x \rightarrow 0} \frac{1}{x} \text{ doesn't exist}$$

vertical asymptotes of the curve  $y = f(x)$ :  $x = a$

$$\text{if either } \lim_{x \rightarrow a^\pm} f(x) = \pm\infty. \quad (\text{Page 95}).$$

§2.3

Limit Laws (Pages 99, 101).

Direct substitution Property:  $\lim_{x \rightarrow a} f(x) = f(a)$  when  $f$  is a polynomial  
or rational function and

$a$  is in the domain of  $f$ .

Squeeze Theorem

$$f(x) \leq g(x) \leq h(x), \quad \lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x) = L$$

$$\Rightarrow \lim_{x \rightarrow a} g(x) = L.$$

§2.4.

Precise definition of a limit.

$$\lim_{x \rightarrow a} f(x) = L \iff \forall \epsilon > 0, \exists \delta > 0 \text{ such that}$$

$$|f(x) - L| < \epsilon \text{ for any } 0 < |a - x| < \delta. \quad (\text{Page 110})$$

Key: How to find a  $\delta$  for a given  $\epsilon$ ?

Similar. Left-hand limit and right-hand limit.

§2.5.

A function  $f$  is continuous at a number  $a$  if  $\lim_{x \rightarrow a} f(x) = f(a)$ ,  
otherwise we say  $f$  is discontinuous at  $a$ . (i.e.  $\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = f(a)$ )

continuous from the left:  $\lim_{x \rightarrow a^-} f(x) = f(a)$

the right:  $\lim_{x \rightarrow a^+} f(x) = f(a)$

continuous on a interval

The following functions are continuous on their domains:

polynomials, rational functions, root functions

(inverse) trigonometric functions, exponential/logarithmic functions

Composite rule

$g$  is continuous at  $a$  and  $f$  is continuous at  $g(a)$ , then  $f \circ g$  is continuous at  $a$ .

The intermediate value Theorem

§2.6

Definition of Limits at Infinity:  $\lim_{x \rightarrow +\infty} f(x) = L$  or  $\lim_{x \rightarrow -\infty} f(x) = L$ . (Page 131)

Horizontal Asymptotes of  $y = f(x)$ :  $y = L$

if  $\lim_{x \rightarrow +\infty} f(x) = L$  or  $\lim_{x \rightarrow -\infty} f(x) = L$

$$\lim_{x \rightarrow +\infty} \frac{1}{x^r} = 0 \quad (r > 0)$$

$$\lim_{x \rightarrow -\infty} \frac{1}{x^r} = 0 \quad (r > 0 \text{ and } x^r \text{ is defined}).$$

$$\lim_{x \rightarrow -\infty} e^x = 0$$

Infinite Limits at Infinity; Prose definitions

§2.7.

The tangent line to  $y = f(x)$  at  $P(a, f(a))$  is given by

$$\frac{y - f(a)}{x - a} = m$$

$$\text{where } m = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

derivative of a function  $f$  at  $a$  is

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} \quad (\text{Page } 146).$$

⇒ The equation for the tangent line

$$y - f(a) = f'(a)(x - a)$$

Position function  $s(t) \rightarrow$  instant velocity  $v(t) = s'(t)$

§2.8.

Derivative function of  $f(x)$ :  $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

$f$  is differentiable at  $a$  if  $f'(a)$  exists at  $a$

$f$  differentiable at  $a \Rightarrow f$  continuous at  $a$ .

Notations:  $y = f(x) \rightarrow y' = f' = \frac{dy}{dx} = \frac{df}{dx} = Df(x)$

Second derivative:  $y'' = f'' = (f')' = \frac{d}{dx} \left( \frac{dy}{dx} \right) = \frac{d^2 y}{dx^2} \Rightarrow \frac{\text{acceleration}}{dt} = v'(t) = s''(t)$

Third derivative:  $y''' = f''' = (f'')' = \frac{d^3 y}{dx^3}$

$n$ -th derivative:  $y^{(n)} = f^{(n)} = \frac{d^n y}{dx^n}$